

Project Information			
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Project Title	Blackjack Royale – A Distributed Online Blackjack Game		
Project Description	This project aims to develop a distributed online implementation of the card game Blackjack. The system uses a client-server architecture consisting of multiple clients (human players) and multiple servers coordinating game sessions, fault tolerance, and synchronization. Players can join Blackjack tables, place bets, draw cards, and interact with the game through lightweight clients. Multiple server instances maintain synchronized game information to ensure reliability and continuous gameplay even if failures occur during a running session. The project focuses on distributed systems concepts such as dynamic discovery, fault tolerance, reconnect handling, synchronization, and leader election while keeping the implementation lightweight and manageable.		

Architectural Model	The system consists of multiple clients and multiple Blackjack server instances. Clients handle user interaction and gameplay actions such as joining tables, placing bets, requesting cards, and receiving game updates. Clients can represent either human players or automated bot players. Server instances coordinate the game logic and maintain the game state including player hands, dealer hands, turn order, bets and balances, and deck management. Each active Blackjack table is managed by one designated Game Master Server. The Game Master coordinates card dealing, validates player actions, manages dealer turns, and synchronizes the current game state with backup servers for redundancy and consistency. Additional server instances remain synchronized and can automatically take over if the current Game Master fails. Communication between clients and servers uses TCP socket communication.
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Dynamic Discovery	If a client wants to join a Blackjack table, it broadcasts a discovery request into the local network. Available server instances respond with connection information and currently active game tables. Clients automatically connect to an available server without requiring hardcoded addresses. Server instances also discover each other through broadcast communication. Newly started servers automatically join the distributed server pool and synchronize their state with existing servers. This allows the system to dynamically adapt to newly joining or disconnected components during runtime.
Fault Tolerance	The system detects crashed or unresponsive clients through heartbeat monitoring and connection timeouts. If a client disconnects during an active game: - the server temporarily preserves the player state - the player may reconnect within a configurable timeout - the game session is restored automatically If the disconnected player does not reconnect within the timeout period, the player may be replaced by a bot or removed from the table. Server instances continuously monitor the health of other servers. If the Game Master Server crashes: - remaining servers detect the failure - a new Game Master is elected - the last synchronized game state is restored - the game session continues without restarting the table This ensures continuous gameplay and minimizes disruption during failures.
Voting	When a new Blackjack table is created or the current Game Master becomes unavailable, the server instances perform a leader election process to select a new Game Master Server. Leader election is implemented using a simplified Bully Algorithm. Each server instance owns a unique numeric identifier. During an election: - servers exchange election messages - the server with the highest active identifier becomes the new Game Master - the elected server broadcasts its leadership status to the remaining servers The Game Master becomes responsible for: - card dealing - validating player actions - dealer behavior - synchronizing game state - managing round progression This coordination mechanism prevents inconsistent game states and ensures that only one authoritative server controls a Blackjack table at any given time.

System Architecture Desin

Include a clear diagram of your system architecture (key components and connections between them). Nächste Seite die Datei lässt mich nichts verschieben



